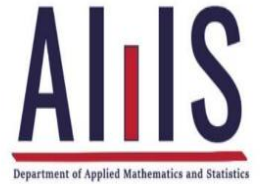




Institute of Technology of Cambodia



Department of Applied Mathematics and Statistics

Group Project: 09

Course Subject: Probability

Topic: Number of rainy days in a month

Probability Report (2025-2026)

Name	ID	Score
1. HIN Panha	e20241363	
2. HENG Lyhour	e20241126	
3. BOEUN Piseth	e20241245	
4. CHHEAN Ravy	e20241154	

Course: Dr. PHAUK Sökkhey

TD: Ms. UANN Sreyvi

Academic year: 2025-2026

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I. Introduction

Weather is an important part of everyday life because it affects transportation, agriculture, business activities, tourism, and people's daily plans. Among different weather conditions, rainfall is one of the most significant factors in Cambodia, especially during the rainy season. Heavy rain can cause traffic congestion, flooding, transportation delays, and difficulties for outdoor activities. Therefore, understanding rainfall patterns is useful for both individuals and organizations.

This project studies the probability of rainy days in Phnom Penh during the month of October 2025. October is usually part of the rainy season in Cambodia, so there are many rainy days during this month. By studying rainfall data, we can estimate how likely it is for rain to occur on a random day in October.

The main problem studied in this project is determining the probability that a day in October will be rainy or dry. To solve this problem, historical weather data was collected and analyzed. The project uses probability concepts to calculate the chance of rain based on observed data from the month.

This problem is important because weather predictions help people prepare for daily activities. Farmers can plan agricultural work, students can prepare for transportation conditions, and businesses can organize outdoor activities more effectively. Rainfall analysis is also important for environmental studies and city planning.

Probability plays an important role in analyzing this problem because it allows us to measure uncertainty. Instead of guessing whether it will rain or not, probability provides a mathematical way to estimate the likelihood of rainfall. By using the number of rainy days and total days in the month, we can calculate the probability of rain and compare it with the probability of dry weather.

Rainfall data in this project from Phnom Penh in October 2025 was used to calculate:

- The probability of a rainy day
- The probability of a dry day

- The percentage of rainy days during the month

The results help us better understand weather patterns in Phnom Penh and show how probability can be applied to real-life situations.

II. Description of the Experiment or Data

The data for this project was collected from historical weather information for Phnom Penh in October 2025.

The observation conducted in this project was counting the number of rainy days and dry days during the month. Each day was recorded as either:

- Rainy day
- Dry day

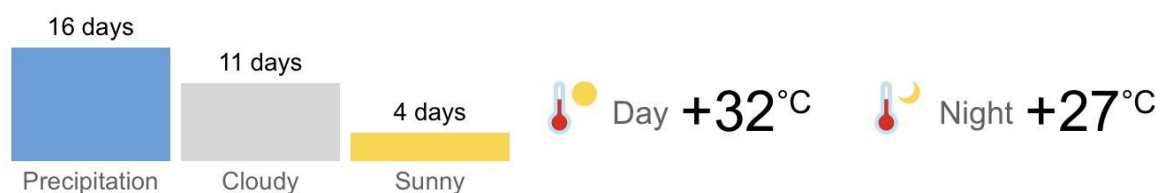
A total of 31 observations were collected because October has 31 days.

According to the data:

- Rainy days = 16 days
- Dry days = 15 days

The data was then used to calculate the probability of rain in Phnom Penh during October.

Average weather in October 2025



III. Define the Random Variable

Let: $X = \text{number of rainy days in October}$

The possible values of X are whole numbers from:

$$0,1,2,3, \dots, 31$$

because the number of rainy days cannot be negative or exceed 31 days.

X is a discrete random variable because it counts the number of rainy days and only takes integer values.

IV. Probability Distribution

A probability distribution table was created based on the rainfall data collected in Phnom Penh during October 2025.

Let:

$$X = \text{number of rainy days in October}$$

The random variable X can take values from:

$$0,1,2,3, \dots, 31 \quad \text{because October has 31 days.}$$

Since data: Total days observed = 31

Rainy days = 16

Dry days = 15

The probability of rain on a given day is: $p = \frac{16}{31} = 0.516$

The probability of a dry day is: $q = 1 - p = 0.484$

The probabilities were calculated using the binomial distribution formula:

$$P(X = x) = C_n^x \cdot p^x (1 - p)^{n-x}, x = 0,1,2,3, \dots, 31, \text{ then}$$

$$P(X = x) = \binom{x}{31} (0.516)^x (0.484)^{31-x} \quad \text{where:} \quad \begin{cases} n \\ p \end{cases} = \begin{cases} 31 \\ 0.516 \end{cases}$$

Probability Distribution Table		
X	<i>Frequency</i>	<i>Probability</i>
0	0.000000005	0.000000000168
1	0.000000173	0.00000000557
2	0.00000276	0.0000000892
3	0.0000285	0.000000919
4	0.000213	0.00000686
5	0.00122	0.0000395
6	0.00567	0.000183
7	0.0216	0.000696
8	0.0690	0.002228
9	0.188	0.006072
10	0.442	0.014250
11	0.899	0.029018
12	1.599	0.051588
13	2.493	0.080425
14	3.419	0.110297
15	4.133	0.133337
16	4.409	0.142226
17	4.150	0.133860
18	3.443	0.111054
19	2.512	0.081050
20	1.608	0.051872
21	0.899	0.028982
22	0.436	0.028982
23	0.182	0.005865
24	0.065	0.002085
25	0.019	0.000623
26	0.005	0.000153
27	0.001	0.000030
28	0.00014	0.0000046
29	0.000016	0.000000509
30	0.0000011	0.000000036
31	0.00000004	0.00000000125

V. Probability Calculations

In this section, probabilities related to the number of rainy days in October are calculated using the probability distribution.

Probability that the number of rainy days is greater than 15

We have to calculate: $P(X > 15)$ or $P(X \geq 16)$

Then $P(X > 15) = P(16) + P(17) + \dots + P(31)$ by cdf

$$P(X > 15) \approx 0.142226 + 0.133860 + 0.111054 + \dots$$

$$P(X > 15) \approx 0.58$$

Therefore: $P(X > 15) \approx 0.58$

There is approximately a 58% chance that October has more than 15 rainy days.

Probability that the number of rainy days is less than or equal to 10

We have to calculate: $P(X \leq 10)$

$$P(X \leq 10) = P(0) + P(1) + \dots + P(10)$$

$$P(X \leq 10) \approx 0.02$$

Therefore : $P(X \leq 10) \approx 0.02$

There is only about a 2% chance that October has 10 or fewer rainy days.

Probability that the number of rainy days is between 14 and 18

We have to calculate: $P(14 \leq X \leq 18)$

Then $\approx P(14) + P(15) + P(16) + P(17) + P(18)$

$$= 0.110297 + 0.133337 + 0.142226 + 0.133860 + 0.111054$$

Therefore: $P(14 \leq X \leq 18) \approx 0.63$

There is approximately a 63% chance that the number of rainy days in October is between 14 and 18 days.

VI. Expect Value $E(X)$

Expect Value represents the average daily rainy day in October.

Since $X \sim \text{Bin}(31, 0.51)$

Then the formula of $E(X) = np$

Thus $E(X) = 31 \times 0.51 = 15.86 \text{ Days}$

The expected value of **15.86 days** represents the average number of rainy days in Phnom Penh during October. This means that if we randomly choose an October month and observe the weather over many years, we can expect rain to occur on approximately **16 days** in that month. Therefore, October can be considered a month with frequent rainfall where rainy days happen nearly half of the month.

VII. Variance and Standard Deviation

- Variance $V(X)$

Variance shows how much the number of rainy days changes from the average rainy days in October. In this project, it explains whether the rainy days are usually close to the expected value or very different from it. **high variance** means the number of rainy days can change a lot from month to month, so some Octobers may have many rainy days while others have much fewer. **low variance** means the number of rainy days stays close to the average, so the rainfall pattern is more consistent every October.

The formula of $V(X) = npq$,

$$(q = 1 - p = 1 - 0.51 = 0.49)$$

Then $V(X) = 31 \times 0.51 \times 0.49 = 7.75$

The variance of **7.75** shows that the number of rainy days in October does not stay exactly the same every year and may vary around the average value. This means the rainy days in October can increase or decrease from the expected **15.86 days** but the variation is still moderate. Therefore, the rainfall pattern in October is fairly consistent, although some months may experience slightly more or fewer rainy days than average.

- Standard Deviation σ

Standard deviation measures how far the data values are from the average value. In this project, it shows how much the number of rainy days in October usually changes from the average number of rainy days (Expect value).

The formula of $\sigma = \sqrt{V(X)}$

Then $\sigma = \sqrt{7.75} = 2.78$

This means the number of rainy days in October usually differs from the average value of 15.86 days by about **2.78 days**. Therefore, most October months are expected to have around 13 to 19 rainy days showing that the rainfall pattern is relatively stable with only moderate variation.

VIII. Conclusion

This project analyzed the probability of rainy days in Phnom Penh during October 2025 using probability concepts and the binomial distribution.

The collected data showed that there were:

- 16 rainy days
- 15 dry days

during the 31 days of October.

The probability of rain on a given day was estimated as:

$$p = \frac{16}{31} \approx 0.516$$

Using this probability, a probability distribution table was created for the number of rainy days in October.

The expected value was calculated as:

$$E(X) \approx 16$$

which means that the average expected number of rainy days in October is about 16 days.

Several probability calculations were also performed. For example:

$$P(X > 15) \approx 0.58$$

This indicates that there is a relatively high chance that October will have more than 15 rainy days.

$$p = \frac{16}{31} \approx 0.516$$

$$p = 3116 \approx 0.516$$

+ Reference:

1. <https://www.weather25.com/asia/cambodia/phnum-penh?page=past-weather#day=17&month=4>

2. https://world-weather.info/forecast/cambodia/phnom_penh/october-2025/

